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Model predictive control for demand flexibility: Real-world operation of a commercial building with photovoltaic and battery systems

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ABSTRACT

Hundreds of studies have investigated Model Predictive Control (MPC) for the optimal operation of building energy systems in the past two decades. However, MPC field tests are still uncommon, especially for small- and medium-sized commercial buildings and for buildings integrated with onsite renewables. This paper describes the implementation and the long-term performance evaluation of an MPC controller in a small commercial building equipped with behind-the-meter photovoltaics and electrochemical batteries. MPC controls space conditioning, commercial refrigeration, and the battery system. We tested two types of demand flexibility applications in the field: electricity bill minimization under time-of-use tariffs and responses to grid flexibility events. Results show that the proposed controller achieves 12% of annual electricity cost savings and 34% peak demand reduction against the baseline, while respecting thermal comfort and food safety. The field tests also demonstrate the ability of the MPC controller to provide a multitude of grid services including real-time pricing, demand limiting, load shedding, load shifting, and load tracking, using the same optimization framework.

1. Introduction

Many countries, including the U.S., have set ambitious decarbonization goals for the next decades [1]. An important strategy to achieve these goals is to make building demand more flexible, which allows the integration of more renewables in the grid and reduces the grid reliance on fossil fuels [2]. To realize demand flexibility (DF), building controls could adapt the building load to be responsive to grid requirements by using local thermal or electrical storage [3]. Several simulation studies have demonstrated the ability of Model Predictive Control (MPC) to optimally aggregate and coordinate multiple flexible loads [4–6]. However, MPC field demonstrations for DF applications are still limited [7,8]. This is particularly true for Small- and Medium-sized commercial Buildings (SMBs) and buildings integrated with onsite renewables.

In the U.S., SMBs represent 50% of the commercial building floor area [9] and making their demand more flexible is paramount for a decarbonized future. Implementing MPC in SMBs requires an approach that is scalable and adaptive to a variety of building energy systems in this building archetype. Unlike large commercial buildings, SMBs are often not equipped with a Building Automation System (BAS) [10]. As a result, any new control software must integrate with diverse hardware and legacy control systems. As highlighted in the literature, the status quo for setting up advanced controls such as MPC requires significant time and effort for engineering and integration [11].

This paper describes a software platform called “SolarPlus Optimizer” (SPO) to address these issues. The platform primarily includes two modules: a middleware and an optimization engine (i.e., an MPC controller). We built the software exclusively on open-source libraries and designed it to be scalable and extensible. We implemented and deployed the platform on a convenience store and gas station with behind-the-meter photovoltaics (PV) and electrochemical batteries. Convenience stores are a good target for control upgrades because they represent over 35% of the brick-and-mortar retail in the U.S. [12]. Advanced controls such as MPC can play an important role in providing demand flexibility to these buildings. In a previous paper [13], we detailed the design, architecture, and deployment of the software platform along with a brief discussion on proof-of-concept experiments.

In this paper, we present the long-term performance evaluation (from September 2020 to August 2021) of the MPC controller using the SPO platform, for various types of scenarios: (1) electricity bill minimization and (2) responses to flexibility events. The first application uses the actual time-of-use (TOU) utility tariff for the test site, including both energy and demand charges. The second one includes five types

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of DF events: Real-Time Pricing (RTP), demand limiting, load shedding, load shifting and load tracking. We then evaluated the performance of the MPC controller against the baseline operation of the energy systems. The paper also documents the controller models used to forecast the system behavior and the generic optimization framework used to handle all the tested scenarios.

We structure the rest of the paper as follows. Section 2.1 reviews the existing literature on MPC, with a focus on experimental and field studies. Based on the literature review, Section 2.2 identifies the research gaps and states the contributions of this paper. In Section 3 we describe the methodology, including details about the building energy systems, field tests design and evaluation methods. Section 4 presents the MPC approach including modeling, optimization, and implementation. Section 5 presents operation results and comparisons with the baseline operation. In Section 6, we detail discussions and recommendations for future work. The paper closes with conclusions in Section 7.

2. Background and contributions

2.1. Related work

Research on Model Predictive Control (MPC) has received substantial interest in the building engineering field in the past two decades [14,15]. This advanced control method has shown great potential for the optimal operation of building energy systems for various objectives: energy efficiency, energy cost savings, CO_2 emission reduction, and thermal comfort improvement as well as a combination of multiple objectives (e.g., in [16–18]). Simulation studies have investigated MPC applications on a wide range of building systems, such as: passive (e.g., building mass) and active thermal storage (e.g., ice storage tanks) [19,20]; Air Handling Units (AHUs) and Variable Air Volume (VAV) Boxes [21–23]; automated windows in mixed natural and mechanical ventilation system [24,25]; as well as coordination of multiple systems (e.g., HVAC, energy storage and PV generation) [4–6]. However, MPC field studies on real buildings are still rare [7,8].

An early study in 2011 implemented an MPC controller on a university building for two months [16,26]: In [16], the authors focused on the subspace method used for the process of model identification, while [26] presented the control details and experimental results. The energy savings of the MPC controller were shown to range from 15 to 28% compared with the existing rule-based control. Issues such as the occurrence of oscillations in the optimal input signal remained to be solved, as discussed in [27]. Corbin et al. [25] proposed an MPC environment for real-time building applications by coupling EnergyPlus and MATLAB; only offline optimization results were however presented due to slow computation speed.

Bengea et al. [28] implemented an MPC controller for a multi-zone VAV system in a research laboratory as a proof of concept. The control variables included mass flow rate, mixed air temperature, supply air temperature and discharge air temperatures. The MPC implementation was carried out along with a pneumatic control retrofit to direct digital control, as well as improvement of local control measures. The overall savings due to control retrofit was 70% on average during the heating season, and the authors mentioned that it was hard to isolate the individual savings due to MPC. Using a patented MPC technology [29], West et al. [17] reported trial results of the software tool on two office buildings, one over two winter months; the other over 10 days. The system showed sound performance for energy reduction without substantially affecting thermal comfort. The study did not present and discuss zone-level controls in detail. Sturzenegger et al. [11] implemented an MPC controller on an office building. The controlled systems are thermally activated building systems (TABS), an AHU system, and window blinds. The MPC strategy was compared to the previously installed industry-standard Rule-Based Control (RBC) strategy for thermal comfort and energy consumption using EnergyPlus simulation results. The authors argued that with current tools available, the operating cost savings alone was difficult to justify the initial cost of the MPC deployment.

Kim et al. [30] tested an MPC-based approach called “unit coordinator” to schedule on/offs of the rooftop units (RTUs) to reduce energy and peak demand in a gymnasium. The team further expanded the MPC approach to include applications in other types of buildings (e.g., offices) with different modeling and optimization methods [31]. The process of data collection and system identification was detailed in another paper [32], where the modeling of disturbance was given special care. An identification method entitled “lumped disturbance” was proposed, and the theoretical analysis was discussed in detail in [33]. The implementation and deployment of the control system relied on commercial software tools such as MATLAB. To test the DF using a TOU tariff, the same team expanded the existing MPC structure to a two-layer MPC formulation [34]. The upper-level MPC responds to the utility price and weather signals by looking at long-term building thermal dynamics, while the lower-level MPC optimally coordinates multiple RTUs to track the optimal setpoints. The lab experiments lasted over 2 months and it showed that the new MPC structure could reduce 30% of demand cost and 4 to 9% of energy cost.

De Coninck and Helsen [18] detailed the practical implementation and evaluation of MPC in an office building. MPC presented up to 40% energy savings compared with the existing RBC strategy. In a Solar Decathlon house, a hybrid MPC [43] was implemented to coordinate natural ventilation through window control, mechanical cooling from a heat pump, Photovoltaic-Thermal (PVT) collectors and thermal storage through Phase Change Materials (PCM). The work focused on comparisons between simulation and experimental results, both of which showed a significant improvement of heat pump Coefficient Of Performance (COP) by using the proposed MPC.

MPC experiments were carried out on a university campus level in [45] by controlling the central chiller system. The model employed in the experiments was improved based on modeling work presented in [50], though the model still could not capture the peak load values in the measurements. Three-week tests showed an improved coefficient of performance of the chilling system and 10% of energy savings. Vrettos et al. [42,51] investigated incorporating MPC of building HVAC systems for frequency regulation as an ancillary service to the grid. A three-tiered control framework was proposed, and the frequency regulation was conducted through varying fan speeds. The control method was implemented and tested in a laboratory environment.

Bruno et al. [40] described a cloud-based architecture to optimally control building loads and photovoltaic and battery systems for demand response. Several control scenarios were mentioned, while only open-loop control was however discussed. Computation time for load forecasting was reported to be several minutes, compared with the control interval of 15 minutes as a limitation. Carli et al. [37] deployed an Internet of Things (IoT) based architecture for MPC of HVAC systems in a one-room laboratory. The paper focused on the communication network and showed field-testing results as a proof-of-concept demonstration of the proposed approach.

Cotrufo et al. [41] implemented a cascaded modeling structure for MPC on an institutional building to reduce natural gas consumption in winter. Black-box models were used to predict the heating demand and electrical baseload, while the indoor air temperature was predicted using a first-order RC model. The algorithm was however only optimizing the transition between night setback and daytime indoor air setpoint temperature. Results showed that 22% of natural gas consumption was reduced during a winter season. Drgona et al. [38] field tested a cloud-based MPC solution for an office building equipped with TABS and geothermal heat pump systems. They used the white-box modeling approach and experimental results showed 53% of heat pump energy savings with a 36% thermal comfort improvement compared with an RBC strategy. On the other hand, despite the accurate predictive capabilities using a high-fidelity model, the authors stated that MPC did not always fully satisfy the defined comfort bounds.

Table 1
Existing studies on the MPC implementation on real building systems: FCU for fan coil unit, RTU for rooftop unit, TABS for thermally activated building systems, PV for photovoltaics, PVT for PV and thermal, PCM for phase change material, AHU for air handling unit, VAV for variable air volume, COP for coefficient of performance, GHG for green house gas.

Year	Ref	Building Type	Test Duration	Systems Controlled	Summary of Results
2022	[35]	University building	2 months	underfloor air distribution system	40% HVAC energy savings
2022	[36]	One office room	10 days	Cooling with a FCU	26–32% energy savings
2022	[34]	Lab testing	2.5 months	RTUs	4–9% energy cost reduction; 30% demand cost reduction
2020	[37]	One test room	4 months	Fan coils	15–20% energy reduction
2020	[38]	Office	6 months	TABS, geothermal heat pump	53% Heat pump energy reduction; 36% thermal comfort improvement
2020	[39]	House	Several days	Heat pump, hot water tanks	2–15% energy cost savings
2019	[40]	School	Weeks	Battery and PV systems	4–7% energy cost savings
2019	[41]	Government building	3 months	Boilers	22% GHG reduction
2018	[31]	Retail store; lab testing	4 months	RTUs	12% energy reduction; 18% demand reduction
2018	[42]	Lab testing	1 week	Fan control for frequency regulation	The facility can track fast-moving signals with high accuracy
2017	[43]	Lab testing	1 week	heat pump, PVT thermal storage (PCM)	Improvement of heat pump COP
2016	[11]	Office	7 months	AHU, TABS	17% energy reduction; 16.9% energy cost savings
2016	[18]	Office	Winter season	Heat pumps, boiler	40% energy reduction; 30% energy cost savings
2014	[44]	One test room	Several days	One fan coil	53% HVAC energy reduction
2013	[17]	Two Offices	2 months	AHU	19% / 32% energy reduction
2013	[28]	Government building	1 week	Dual-duct VAV	70% energy reduction; 10% demand reduction
2012	[45]	University campus	5 days	District cooling (chillers and ice storage)	11.9% COP improvement
2012	[46]	School classroom	2 days	Air conditioner	30–70% energy reductions
2012	[47]	School classrooms	Several days	AHU	50% energy reduction
2011	[26]	University building	2 months	TABS	15–28% heating energy reductions
2011	[48]	Two offices	2 months	Shading; window opening	Thermal comfort improvement
2010	[49]	Office	40 days	Boiler	Thermal comfort improvement
2009	[50]	University campus	3 weeks	District cooling (chillers and ice storage)	10% energy reductions
2005	[20]	Two test rooms	4 days	Chillers, ice storage	17–27% energy cost savings

A machine learning based MPC was tested with a fan coil unit (FCU) for a 1-zone office located in the basement of a building [36]. Two types of MPC approaches were compared: one with linearization and one without. The 10-day tests showed that the two approaches reduced energy by 26% and 32%, respectively. Disturbances from weather were not considered due to the unique location of the studied office. Blum et al. [35] integrated an MPC controller into the existing BAS on a large office building. The controlled HVAC system is an underfloor air distribution system with RTUs providing cooling. The two-month tests during a transition season showed that MPC could reduce 40% of HVAC energy without significant penalty to thermal comfort.

2.2. Research gaps and contributions

Table 1 summarizes all the work so far on MPC implementations on real systems as to the authors' knowledge. We can see that the scope of MPC field studies has been limited to a small subset of buildings, mostly office spaces, laboratories, experimental testbeds, or university buildings. Experiments were often conducted in well-instrumented environments, which included more measurements with more variables than general buildings. In those studies, most relied on proprietary software coupled with the existing BAS system in the building. Few studies investigate SMBs, likely due to the lack of BAS to control the various energy systems. Another observation is that most existing papers either focus on HVAC systems, typically implemented by building or mechanical engineers, or PV and battery systems, typically implemented by electrical engineers. Very few studies investigated the combination of HVAC, refrigeration, and DERs and the coordination and aggregation of these sub-loads at a supervisory level. In addition, field studies on the DF of buildings are rare, and energy efficiency still dominates as the goal of field investigations. With the trend of building electrification and grid-interactive buildings, the DF of buildings will become more and more important, and call for more field validation work.

Given these gaps in the literature, the goal of this paper is to implement and operate an MPC controller using SPO on an SMB to achieve DF, as well as evaluate its performance against a baseline control strategy. The contributions of the paper include the following:

- The real-world operation of the MPC strategy for the DF of an SMB integrated with onsite renewables contributes a valuable and early case study to the literature. The operating results demonstrate the flexibility potential of such a building archetype and provide evidence for the wider adoption of DF.
- The tests conducted show that the MPC strategy can optimally coordinate HVAC, refrigeration, and DERs to achieve the desired flexibility goals. This is unique in the literature, where authors focus on either thermal systems (i.e., HVAC, thermal storage, and refrigeration) or DERs.
- The presented optimization framework can handle different grid signals with the same formulation: five grid-interactive DF modes as well as bill optimization (i.e., energy and demand cost minimization). This capability is novel and has not been demonstrated in real buildings before.
- The work presented in this paper is a demonstration of the capabilities of the SPO platform, built solely on open-source software. The system is also capable of reliably operating long-term and in various conditions.

3. Methodology

3.1. System description

The building selected for the field demonstration of the new controller is a convenience store, with an associated refueling station, located in Blue Lake Rancheria, California. The work described in this paper is part of a project to transform this site into a small-scale microgrid providing refueling and cold storage services to this rural community [52]. The increased frequency of Public Safety Power Shutoffs (PSPS) (i.e., planned power outages at times of high risk of wildfires) in this area has made the need for a resiliency hub more evident. The controller will therefore play an important role in optimizing energy use and improving the overall resiliency provided by the system [53].

The main space inside the store is conditioned by two separate RTUs, where one unit serves the east zone and the other serves the west zone, including an enclosed office and a restroom. No physical separations, e.g., partition walls, exist between the open areas of the two zones. In

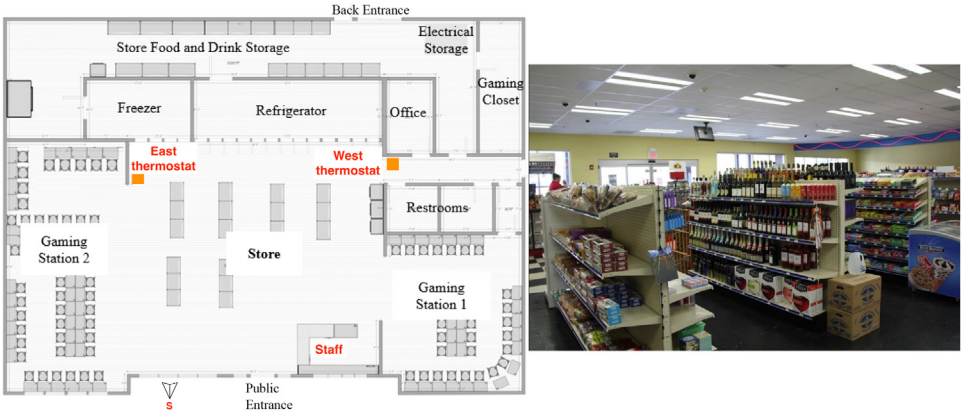


Fig. 1. The layout of the store with thermostat locations (left); main space for dry food display (right). The public entrance and all windows are south facing.

Table 2
Nominal or peak power of the building and its subsystems studied in this work.

System	Nominal or peak power (kW)	Number
RTU	9.6	2
Refrigerator	4.5	1
Freezer	3.6	1
Battery	±109	1
PV	- 60	1
Building	49	-

the store, the major heat source is from the gaming machines, part of the adjacent casino in the area. They are not allowed to stop or restart according to regulations. The two RTUs are controlled by individual thermostats that have been installed in the corresponding zones (see Fig. 1). As shown in the layout, in the middle of the store, there is a refrigerator room for storing and displaying beverages with glass reach-through doors and a freezer room for storing and displaying ice cream and frozen food. On the backside of the store, the storage room for food and drink is not conditioned.

As part of the project, the site has installed PV panels on the gas station canopy with 60 kW capacity, along with a battery system with a storage capacity of 174 kWh and peak discharge power of 109 kW. Both the PV generation and battery systems are oversized compared with the peak demand of around 49 kW of the store. The project team made this choice to consider future integration with the microgrid system of the community and to provide resiliency to power outages due to the PSPS events [53]. Table 2 summarizes the peak building load and its energy system nominal or peak power. Note that the negative sign for the PV indicates that it is the local generation, and the positive-negative sign of the battery indicates the maximum charging and discharging rate. The store is typically open 24 h a day, 7 days a week, and serves customers throughout the day.

3.2. Business-as-usual control

In Business-as-usual (BAU) operation, the facility operator sets the RTUs, refrigerator, and freezer systems to operate at a constant setpoint without any setback as the store is occupied 24/7. The defrost cycles of the refrigerator and freezer systems are controlled with a timer. The battery is controlled using advanced commercial software purchased from the battery vendor. The software is capable to optimize the electricity cost by adjusting the battery charging and discharging behavior based on the TOU tariff and onsite PV generation. Its algorithm is however proprietary and unknown to users. Table 3 summarized the setpoints of the BAU operation mode. To evaluate the SPO performance, we set this BAU mode as the baseline control strategy.

3.3. Field tests design

We planned to test two types of DF applications: electricity bill minimization using Time-of-Use (TOU) electricity tariffs and event-based DFs. The first test evaluates the long-term performance of SPO using the current building tariff, while the latter evaluates SPO's capability to respond to future-looking scenarios of grid signals. We explored five types of grid signals in this work including Real-Time Pricing (RTP), demand limiting, load shedding, load shifting, and load tracking. The following subsections detail the evaluation criteria and test plans for each test.

3.3.1. Bill minimization

The electricity bill for commercial buildings in California generally includes two variable parts related to the electricity usage: the cost for time-of-use energy and the cost for peak demand. The demand is normally charged based on the maximum power each month, sampled every 15 min. The contract of the facility owner with the local utility uses a more advanced tariff structure, which includes: (1) a time-of-use demand charge based on the highest demand in each of the time-of-use periods (peak, mid-peak, and off-peak) with a cost coefficient that varies every season (winter, spring, summer); (2) a time-of-use cost for the energy component of the bill [54]. The winter season lasts for five months from October to February, the spring season lasts for three months from March to May, and the summer season lasts for four months from June to September. In winter, a peak period is defined as 16:00–21:00, and the off-peak period lasts for the rest of the day. The differences in energy and demand prices between peak and off-peak periods are relatively low compared to those in summer. In spring, the rate includes a super-off-peak period (9:00–14:00), when the energy rate is even lower than the winter off-peak rate. In summer, a part-peak period is added between peak and off-peak, and the demand charge for the peak period is significantly larger than the off-peak period. Table 4 summarizes the details of the tariff. Note that an additional monthly demand charge of $21.08/kW for the maximum demand of the month applies to each month throughout the year. Eqs. (1) and (2) show the KPIs for the bill minimization, which are defined as the absolute electricity cost savings and the percentage of savings.

$$\$_{savings} = \$_{BAU} - \$_{SPO} \tag{1}$$

$$\$_{savings}\% = \frac{\$_{BAU} - \$_{SPO}}{\$_{BAU}} \times 100\% \tag{2}$$

where $\$_{BAU}$ is the electricity cost under the BAU mode, while $\$_{SPO}$ is the cost under the SPO mode.

We tested the scenario of electricity bill minimization for each of the three seasonal tariff variations of B19 from September 2020 to August 2021.

Table 3
Setpoints or setpoint ranges of the controlled energy systems in the two operation modes: Business As Usual (BAU) and SolarPlus Optimizer(SPO).

Operation mode	RTUs setpoint (°C)	Fridge setpoint (°C)	Freezer setpoint (°C)	Battery power (kW)
BAU	21.1	-0.56	-21.7	Proprietary
SPO	[19.5, 22.6]	<=0	<=-18	[-27, 27]

Table 4
The electricity tariff B19 of the California utility PG&E tested in this work.

		Peak	Part Peak	Off Peak	Super Off Peak
Winter (Oct-Feb)	Period	16:00–21:00		Rest of day	
	Energy rate ($/kWh)	0.14		0.11	
	Demand rate ($/kW)	1.79		0	
Spring (Mar-May)	Period	16:00–21:00		Rest of day	9:00-14:00
	Energy rate ($/kWh)	0.14		0.11	0.068
	Demand rate ($/kW)	1.79		0	0
Summer (Jun-Sep)	Period	12:00–18:00	8:30–12:00; 18:00–21:30	Rest of day	
	Energy rate ($/kWh)	0.16	0.13	0.11	
	Demand rate ($/kW)	25.58	5.23	0	

3.3.2. Real-time pricing

RTP is a price-based flexibility signal, which contains a dynamic energy price at each sampling rate (e.g., every 15 min) and no demand charge. The RTP signal tested in the study aims to counterbalance the duck-curve shape of the grid operated by the California Independent System Operator (CAISO) [55]. On sunny days, the grid experiences a low demand in the midday and a steep ramp to peak demand in the evening due to the high penetration of solar power generation. This price signal was generated based on K-means clustering of historical day-ahead wholesale prices in California, which is a camel-shape curve [13]. It was then scaled for bill neutrality with the TOU rate. The KPI for the RTP test is the same as Eq. (2). However, the electricity cost for both BAU and SPO in this test is simply the product of net energy use and RTP.

3.3.3. Demand limiting

We define the demand limiting event as limiting the building demand below a certain upper bound P_{lim} within a certain period between start time t_s and final time t_f. Eq. (3) shows the KPI to evaluate this event, which denotes the percentage of time that the building demand is below the given limit.

$$time\% = \frac{\int_{t_s}^{t_f} t_{P_{SPO}<P_{lim}}}{\int_{t_s}^{t_f} t} \times 100\% \tag{3}$$

where $t_{P_{SPO}<P_{lim}}$ denotes the time when the demand in the SPO mode is below the given demand limit P_{lim}.

In the demand limiting test, we defined the test time window from 15:00 to 18:00 and set the building net load to be lower than 26 kW during the event.

3.3.4. Load shedding

In a load shedding event, the grid asks the building to reduce a certain kW of demand P_{dec} from its normal operation during a certain period t_{dec}. It is a variation of the demand limiting event. In a demand limiting event, the grid simply imposes a cap on the building, while in a load shedding event, the grid requires the building to reduce a certain amount of demand based on its historical demand P_{ref}. Eq. (4) presents the KPI used to evaluate the load shedding event. It denotes the percentage of time when the demand under the SPO mode satisfies the given amount of load reduction during the event.

$$time\% = \frac{\int_{t_s}^{t_f} t_{P_{SPO}<(P_{ref}-P_{dec})}}{\int_{t_s}^{t_f} t_{dec}} \times 100\% \tag{4}$$

where $t_{P_{SPO}<(P_{ref}-P_{dec})}$ indicates the time when the demand under the SPO mode is below the reference power minus the required power reduction.

We defined the time windows for this test the same as the demand limiting event, from 15:00 to 18:00. In this test, we considered a situation where the battery was controlled at a reduced maximum charge/discharge rate so that we could study the ability of the optimization engine to control the RTU and refrigeration systems in response to the grid signal. The required amount of the load reduction is 3 kW against the BAU operation during the test period.

3.3.5. Load shifting

In a load shifting event, the grid requests the building to shift load from a peak period to a non-peak period. This event can be considered a load shedding event plus a load-increase event. During the peak period, the effect of power reduction is the same as that in the load shedding event. During the off-peak period, the building also needs to increase a certain kW of power P_{inc} from its normal operation during the load-increase period t_{inc}. Eq. (5) shows the KPI to evaluate the load-increase period, which is the percentage of time when the demand under the SPO mode satisfies the amount of load increase. Eq. (4) can be used to evaluate the load-decrease period.

$$time\% = \frac{\int_{t_s}^{t_f} t_{P_{SPO}>(P_{ref}+P_{inc})}}{\int_{t_s}^{t_f} t_{inc}} \times 100\% \tag{5}$$

where $t_{P_{SPO}>(P_{ref}+P_{inc})}$ indicates the time when the demand under the SPO mode is above the reference power plus the required power increase.

In the load shifting test, we defined a load-decrease period from 16:00 to 21:00 and a load-increase period from 9:00 to 14:00. The building was required to consume 10 kW more against its BAU operation during the load-increase period and to consume 10 kW less during the load-decrease period.

3.3.6. Load tracking

During a load tracking event, the grid requests the building to follow a reference power profile P_{ref} within an allowed margin of error ϵ. Eq. (6) presents the KPI to evaluate the load tracking event, which is the percentage of time when the building load does not deviate from the reference power profile within the given error band.

$$time\% = \frac{\int_{t_s}^{t_f} t_{|P_{SPO}-P_{ref}|<\epsilon}}{\int_{t_s}^{t_f} t} \times 100\% \tag{6}$$

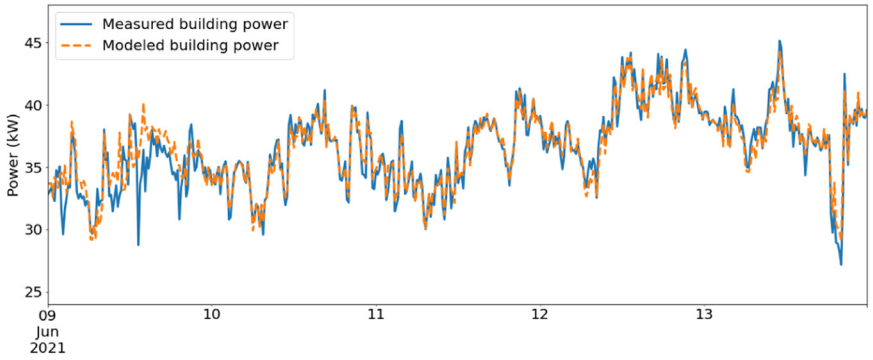


Fig. 2. Model validation results of the total building power for the baseline operation mode Business As Usual (BAU) for five consecutive days.

where $t_{|P-P_{ref}|<\epsilon}$ indicates the time when the difference between the building load and the reference power profile is within the given error band.

In our test, we used a flattened building load profile compared to its normal operation as the grid signal. This signal also aims to counterbalance the duck-curve behavior of the CAISO grid. The allowable error of margin during this 24 h test was ±3 kW.

3.4. Business-as-usual baseline

To evaluate the performance of the SPO operation for the bill minimization and the RTP test, we compare its operation with the BAU mode as described in Section 3.3. Using only the operational data under the BAU mode, we created a baseline regression model, which aims at predicting the building loads of BAU under the same weather conditions as SPO according to standard measurement and verification practice [56]. This baseline load model uses a black-box Random Forest (RF) regression algorithm [57]. The total building power was regressed as a function of outdoor dry bulb temperature, solar irradiance, day of the week, time of the day, and the power of uncontrolled loads. We then validated the model based on the eight-fold cross-validation method [58]. Over the whole validation period, the model achieves an R^2 value of 0.91 and a Root Mean Square Error (RMSE) of 1.9 kW. Fig. 2 shows the total building power predictions compared to the measurements during five weekdays. We can observe that the model predicts the total building power accurately most of the time. Using this baseline model, we then obtained the extrapolated total building power of BAU under the same weather conditions as SPO. The battery and PV power data from the BAU operation were added to this extrapolated building power to get the building net power of BAU.

4. Model predictive control

4.1. Modeling

4.1.1. Building systems

The layout of the building spaces is shown in Fig. 1 as described in Section 3.1. The RTU on the east side serves the east part of the store, while the RTU on the west side serves the west part of the store, the office and the restroom. The two RTU systems are controlled independently by two thermostats. The storage room and gaming closet on the north side are not conditioned. The refrigerator room is cooled by a refrigerating system, while the freezer room is cooled by a freezer system. The refrigerating and freezer systems are independently controlled by their own controllers. To represent the layout of the conditioned spaces, the building thermal model thus consists of four zones: east, west, refrigerator and freezer zone.

The building thermal zones are modeled using the lumped resistance and capacitance approach. In the employed model, each zone is represented by one resistance and one capacitance. This model structure

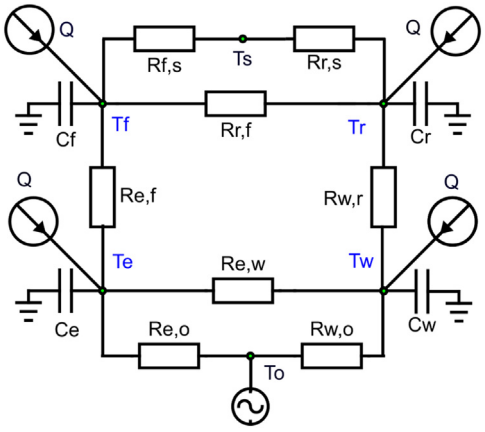


Fig. 3. The Resistance-Capacitance (RC) network of the building thermal model: R indicates the resistance between the connected temperature nodes; C indicates the capacitance; Q indicates the net heat addition to the nodes.

could capture the thermal dynamics without being too complex, as commonly used in control-oriented energy models [59–61]. Fig. 3 shows the thermal connections of the nodes, including the outdoor air temperature $T_o(t)$. In the figure, $T(t)$ indicates the air temperature of each node, where the four zonal nodes $T_e(t)$, $T_w(t)$, $T_r(t)$, $T_f(t)$ represent respectively the east, west, refrigerator, and freezer zones. $T_s(t)$ represents the temperature of the storage room, which is unconditioned and uncontrolled. R represents the lumped thermal resistance between two nodes and C represents the lumped thermal capacitance of the four zones with thermal dynamics. $Q(t)$ in each zone indicates all the active and passive heat gains and losses including solar heat gains $Q_s(t)$, internal heat gains $Q_i(t)$, heat injection $Q_h(t)$, and subtraction $Q_c(t)$ from mechanical systems.

Mathematically, the building thermal model can be written using Eqs. (7)–(10). The overall building thermal model is therefore a fourth-order linear model, which allows efficient computation for the optimization algorithm.

$$C_e\dot{T}_e(t) = \frac{T_e(t)-T_o(t)}{R_{e,o}} + \frac{T_e(t)-T_w(t)}{R_{e,w}} + \frac{T_e(t)-T_f(t)}{R_{e,f}} + Q_{ce}(t) + Q_{he}(t) + Q_{ie}(t) + Q_{se}(t) \tag{7}$$

$$C_w\dot{T}_w(t) = \frac{T_w(t)-T_o(t)}{R_{w,o}} + \frac{T_w(t)-T_e(t)}{R_{e,w}} + \frac{T_w(t)-T_r(t)}{R_{w,r}} + Q_{cw}(t) + Q_{hw}(t) + Q_{iw}(t) + Q_{sw}(t) \tag{8}$$

$$C_r\dot{T}_r(t) = \frac{T_r(t)-T_f(t)}{R_{r,f}} + \frac{T_r(t)-T_w(t)}{R_{w,r}} + \frac{T_r(t)-T_s(t)}{R_{r,s}} + Q_{cr}(t) \tag{9}$$

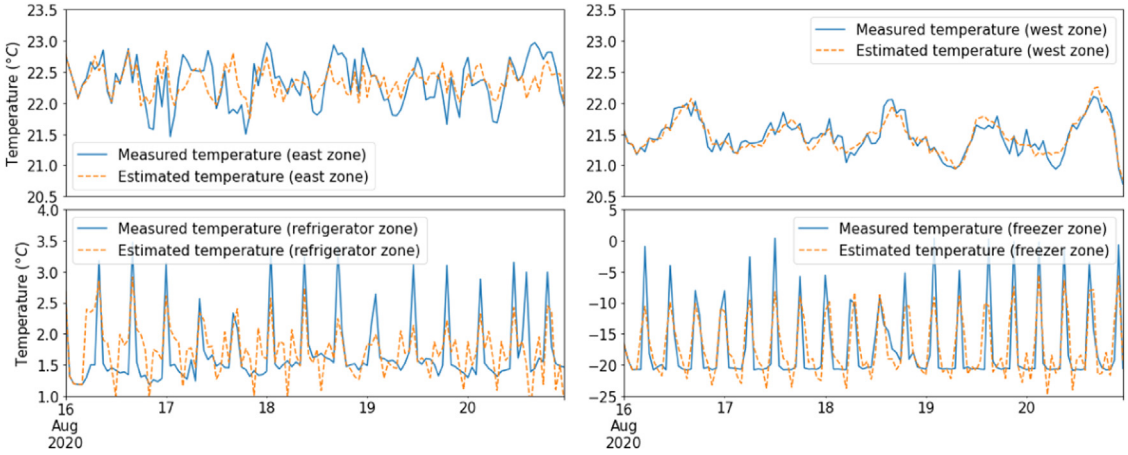


Fig. 4. Model validation results of the building thermal model using data from the baseline operation mode Business As Usual (BAU) for five consecutive days.

$$C_f \dot{T}_f(t) = \frac{T_f(t) - T_r(t)}{R_{r,f}} + \frac{T_f(t) - T_e(t)}{R_{e,f}} + \frac{T_f(t) - T_s(t)}{R_{f,s}} + Q_{cf}(t) + Q_{hf}(t) \tag{10}$$

The nominal heating and cooling capacities of the RTU systems are, respectively 29.3 kW and 24.91 kW. The rated heating efficiency is 0.8 and the rated cooling COP is 2.59 for each RTU. The refrigerator has a nominal cooling capacity of 5.861 kW and a rated COP of 1.3. The freezer has a nominal cooling capacity of 6.096 kW with a rated COP of 1.7 and a nominal defrost capacity of 4.5 kW. Due to a lack of manufacturer data and unmeasured heat flow from these mechanical systems, it was difficult to construct performance-map-based models for the mechanical systems. We therefore used their nominal capacity and efficiency or COP in the model for the HVAC&R systems.

The PV array provides a nominal 60 kW direct current peak power. Note that the PV is not directly controlled by the MPC controller. A steady-state model was used for the PV system, where the solar radiations on the PV surfaces were forecasted as described in the section below. The battery system provides 109 kW peak power and the total energy capacity is 174 kWh. The battery is modeled using the bucket model approach by considering the battery as a repository for energy [62]. The model can be written using Eq. (11).

$$\dot{SOC}(t) = \frac{P(t)}{E} \tag{11}$$

where SOC indicates the State Of Charge (SOC) of the battery, P the charging or discharging power, and E the rated energy capacity.

We estimated the RC parameters of the building thermal model using the operation data under the BAU mode. We then calibrated the model using a different dataset. Fig. 4 presents validation results of the measured and estimated temperatures of the four thermal zones for five days. We can see that globally, the estimated temperatures capture the trend of the measured temperatures quite well. The Coefficients of Variation of the Root Mean Square Error (CVRMSE) of the four zones are respectively 1.4% for the east zone, 0.6% for the west zone, 25.9% for the refrigerator zone, and 22.2% for the freezer zone, which are all smaller than 30% as required by ASHRAE Guideline 14 [63] for hourly data. The refrigerator and freezer zones presented higher errors due to temperature increase in the defrost cycle. The temperature errors were much smaller when the refrigeration systems were in the refrigeration process.

4.1.2. Disturbances

Weather is the predominant external disturbance of the building thermal load, with outdoor dry bulb temperature and solar radiation being the major driving variables. In this study, we forecasted these two variables and incorporated them into the controller model. We obtained the weather forecast data from a third-party application called DarkSky [64], which provides forecasts for outdoor dry bulb temperature, relative humidity, wind speed, and cloud cover. Because the solar radiation data is not directly available from this source, we incorporated a solar radiation model in the controller [65]. The model predicts hourly global horizontal solar radiation from the solar altitude angle, cloud cover, dry bulb temperature, relative humidity, and wind speed. Fig. 5 presents validation results of the global horizontal radiation using the solar model for five days. We can see that the estimated solar radiation matches the measured data well.

After obtaining the global horizontal solar radiation, we split the direct and diffuse solar radiation from the global horizontal radiation based on an approach described in [66]. We then calculated the solar radiation on any titled surface using the solar azimuth angle, the azimuth and titled angle of the object, and ground reflectivity. We conducted this calculation separately for the building and the PV. The overall solar heat gains Q_s of the building were approximated by considering only the solar gains through the windows, which are all south facing as shown in Fig. 1. The solar heat gains were then split into the east and west zones as Q_{se} and Q_{sw} based on the window sizes of these two zones.

Another important source of disturbances is the internal heat gains. As mentioned in Section 3.1, the building is open to customers 24 hours per day, 7 days per week. The major contributors to internal heat gains are the gaming machines, operating 24/7, which inject heat directly to the east and west zones. Occupants in the building are typically made up of 1–3 staff members and a variable number of customers who spend a short time inside the store. The contributions to the internal gains due to people are negligible compared to the gains due to lighting and plug loads (below 1 kW vs. over 20 kW). The internal gains due to occupants are therefore not included in the model. The lighting and plug loads are not sub-metered separately, and it is thus difficult to precisely estimate their contribution to the internal gains. In the model, we assumed the total internal heat gains all from the uncontrolled and unmeasured lighting and plug loads, which were calculated from the total building power (excluding the battery and PV power) minus the total power of the sub-metered HVAC&R systems. Due to the small demand variations of the lighting and plug loads, we used a constant average power of these loads over each season to predict the internal heat gains Q_i. We updated this constant average power in the controller model if we observed a significant change from one season to the other based on the measurements and calculations. Finally, we split the resulting internal gains into the east and west zones as Q_{ie} and Q_{iw} based on floor areas.

In summary, we considered the three disturbance variables (outdoor air temperature, solar radiation and internal heat gains) as known and deterministic signals in this work. We incorporated them into the con-

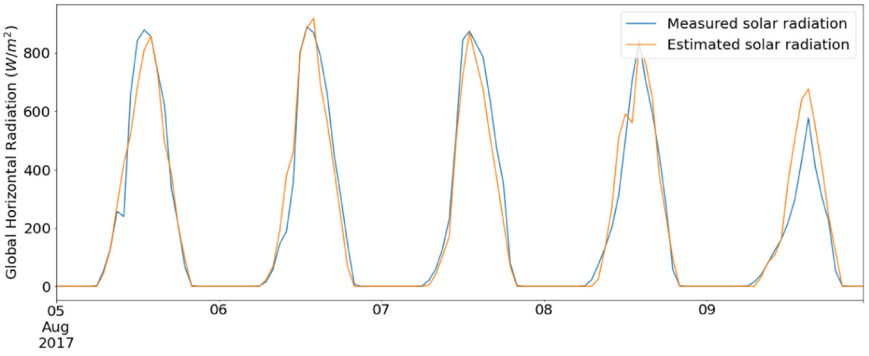


Fig. 5. Model validation results of the solar radiation model for five consecutive days.

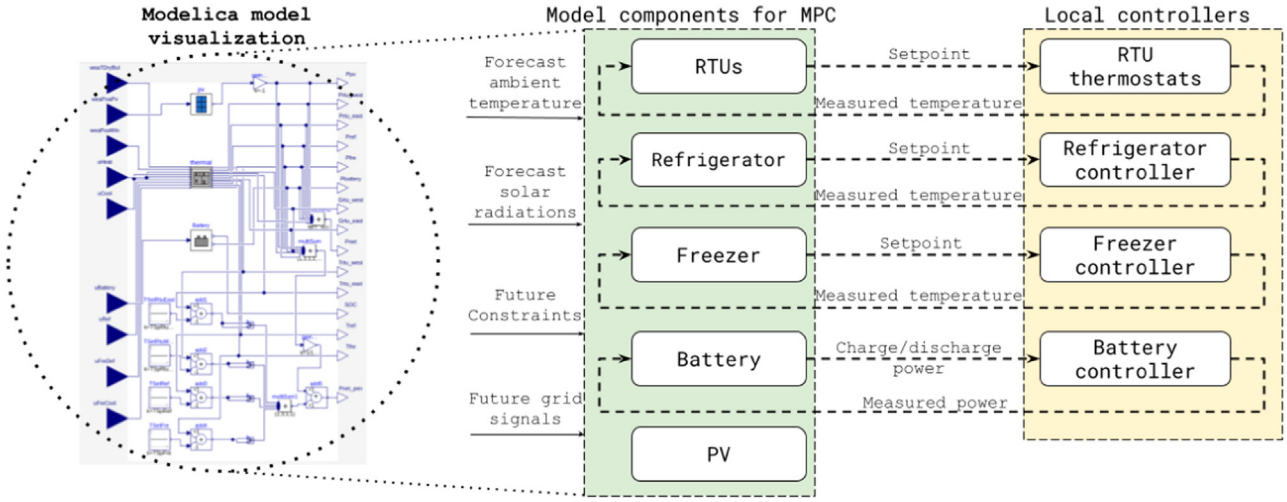


Fig. 6. The communication structure between the supervisory controller and local controllers: the green box in the middle indicates the supervisory controller hosted in the computer; the yellow box on the right indicates the physical controllers [13].

Table 5
Summary of the inputs and outputs of the MPC supervisory controller used in communication with the physical systems.

	Variable type	Variable description
Inputs	Controlled inputs	Temperatures (east zone, west zone, refrigerator, freezer), Battery power
	Uncontrolled inputs	Outdoor temperature, global horizontal solar radiation, internal heat gains
Outputs		Temperature setpoints (east zone, west zone, refrigerator, freezer), Battery charge/discharge power

troller model as uncontrolled inputs and forecasted over the control horizon at each control interval (every 5 min).

4.2. Control system design

The SPO optimization engine (i.e., MPC controller) is designed as a supervisory controller communicating setpoints to the local controllers and reading state variables from them (Fig. 6). The local controllers use these setpoints to control the underlying energy systems (i.e., RTUs, refrigerator, freezer, battery) based on their internal control logic (e.g, turning on and off or changing the equipment stages). Table 5 summarized the inputs and outputs of the MPC supervisory controller. When the energy systems operate under the BAU mode, the setpoints are the default setpoints in their corresponding controller, as shown in Table 3. When SPO is in operation, these setpoints are overridden by the optimal setpoints calculated by the optimization engine. The local controllers feed the measured zone temperatures and the battery power (see Fig. 6) back to the SPO optimizer. With this solution, the energy systems can operate with or without the SPO platform. The setpoints of the two RTU thermostats can also be overridden by the store occupants using the thermostat interface. No additional sensors were installed in each thermal zone, and all the state variables were collected by the local controllers.

The state variables $x = [T_e, T_w, T_r, T_f, SOC]$ used by the SPO optimization engine, and listed in Eqs. (7) to (11) are measured (note that the measured battery power is converted to SOC within the model). No state observers or estimators are therefore needed. The uncontrolled temperature T_s is considered constant. The other uncontrolled input signals w are the forecasted disturbances $[T_o, Q_s, Q_i]$ over the control horizon. The output signals from the optimization engine are the demand and future temperature trajectories. The temperatures are served as the reference setpoints for the local controllers. At each control interval (every 5 min), the controller receives the measurements from the local controllers and updates the controller model states and inputs including the forecasted disturbances, constraints, and incentive signals. It then produces and sends the reference setpoint trajectories to the local controllers and restarts the process for the next control cycle.

4.3. Optimization

4.3.1. Optimization problem formulation

Based on the tariff structure described in Table 4, a monthly electricity bill can be expressed using Eq. (12).

$$Bill = \sum_{m_{start}}^{m_{end}} E(t) \cdot r_e(t) + \sum_{k=1}^{N} P_{max}(k) \cdot r_p(k) + P_{max,m} * r_{p,m} \tag{12}$$

where $E(t)$ denotes the time-varying energy usage that is charged from the start of the month m_{start} to the end of the month m_{end}; $r_e(t)$ and $r_p(k)$ denote separately the time-of-use energy rate and time-of-use demand rate; $P_{max}(k)$ denotes the maximum power at the sampling interval during a specific time-of-use period k; $P_{max,m}$ denotes the monthly maximum power at the sampling interval and $r_{p,m}$ denotes the fixed demand charge rate for this monthly maximum power. The value of N is an integer specifying the number of time-of-use periods in the tariff. In our tested tariff, there are two periods in winter and three periods in spring and summer. Without loss of generality, Eq. (12) can be simplified with only the first and third terms on the right-hand side for a more common tariff structure. The net power P at any time for our building can be expressed in Eq. (13).

$$P = P_{building} + P_{battery,charge} - P_{battery,discharge} - P_{PV} \tag{13}$$

where $P_{building}$ denotes the building total load without the DERs; $P_{battery,charge}$ and $P_{battery,discharge}$ denote separately the charging and discharging power; P_{PV} denotes the PV generation power. These power values on the right-hand side are all non-negative.

To handle the issue of monthly demand charge in a shorter optimization horizon, we formulated the optimization problem in Eq. (14) [67] subject to constraints in Eqs. (15)–(18).

$$J = \min \left[\int_{t_0}^{t_h} P(t,x,u,w) \cdot C_e(t) dt + \sum_{k=1}^{N} \Delta P(t,x,u,w,k) \cdot C_p(k) \right] \tag{14}$$

s.t.

$$\Delta P(t,x,u,w,k) \geq 0 \tag{15}$$

$$\Delta P(t,x,u,w,k) \geq P(t,x,u,w) - P_{th}(k) \tag{16}$$

$$\dot{x} = f(t,x,u,w) \tag{17}$$

$$g_{i,lb}(t) \leq g_i(\cdot) \leq g_{i,ub}(t) \quad \forall i \in I, \tag{18}$$

where we define the optimization horizon from t_0 to t_h, which is set 24 h in our operation. The first integral term is the energy cost during the horizon. The second term denotes the incremental demand cost of the increased power during the horizon compared to a power threshold, $P_{th}(k)$, defined as the maximum power that occurred previously within the month. Eqs. (15) and (16) indicate that the incremental power must be non-negative and larger than the difference between the current power and the threshold power. This approach considers that in each optimization horizon, the demand charge term matters only if it exceeds the highest peak experienced from the beginning of the month (i.e., billing period) up to now.

Eq. (17) denotes the equality constraints with respect to the states x and inputs u defined in Eqs. (7) to (11). Eq. (18) denotes the inequality constraints for the states x, inputs u and outputs P, the collection of which are indicated by I. The constraints for the inputs u are within the physical capacities of the energy systems. Table 3 summarizes the operating ranges for the test building, which is defined by the facility operator. The determined setpoint range for the RTUs considers the acceptable thermal comfort for the occupants; the determined ranges for the refrigerator and the freezer consider the food safety, while the charge and discharge range for the battery considers a reasonable threshold as per the magnitude of the building load. Extra constraints for the outputs (i.e., demand) are shown in the last column of Table 6 to handle the requirements of the various flexibility events with different grid signals.

4.3.2. Handling grid signals

Table 6 summarizes the DF events with various grid signals handled by the optimization engine. The basic grid signal is the TOU tariff for both energy and demand from the utility, which is included in demand limiting, load shedding, and load shifting. In those events, the objective function is the same, which is the minimization of the cost of energy and demand. For these three events, we only need to reformulate the grid requirements as a constraint to the demand, which is shown in the last column of the table. The duration of these signals is generally denoted by the start time t_s and final time t_f, except that the load shifting event is separated into two sections: the period of load increase and the period of load decrease. P_{ref} denotes a time-varying reference power, which may be the site's historical demand, as in the load shedding and shifting signals, or desired building load shape, as in load tracking. For the RTP event, the objective function is to minimize the energy cost excluding the demand charge. For the load tracking event, the objective is to minimize the error between the predicted building power and the given reference power over the control horizon. The margin of error is translated into constraints for the optimization problem. Note that all constraints are implemented as soft constraints in this work (i.e., added as weighted penalties to the cost function).

4.4. Software implementation

Like most SMBs, the selected convenience store does not have a BAS system. The SPO middleware is used to tie together the hardware and the optimization engine. Unlike the proprietary BAS systems in the market, We built SPO exclusively on open-source libraries and designed it to be scalable and extensible, as well as vendor- and protocol-agnostic. In such a way, building managers can take advantage of a larger market of connected devices (both sensors and actuators) without being tied down to any particular manufacturer ecosystem. It also enables less costly adaptation and modification of the system over the expected lifetime of hardware.

The major component of the middleware is the eXtensible Building Operating System (XBOS), an open-source system developed for real-time data acquisition and control of actuators [68]. XBOS consists of WAVE and WAVEMQ [69] for data exchange, drivers for connection to hardware, data storage, and applications (e.g., MPC is one of the applications in our current work). It enables cloud communication and remote control, and it is robust for network outages, where the local system can continue operating without the Internet. Fig. 7 presents the software architecture of the platform, and details about each component can be found in [13].

We implement the optimization engine in the SPO platform using the Python package MPCPy [70]. The package includes generic functionalities such as parameter estimation and constrained optimization. It allows easy testing for different objectives: e.g., minimizing energy, energy cost, both energy and demand costs, as well as for different demand management scenarios.

For modeling, MPCPy relies on models defined in the Modelica language [71], an equation-based multi-domain language, to model complex physical systems including mechanical, electrical, and deterministic control systems. These models can be used to predict future system behavior and to support linked simulation-optimization problems. For use in optimization, MPCPy focuses on the use of simplified physical models, often known as gray-box models. These gray-box models include sufficient information about the building systems to simulate overall responses, without excessive details. The goal of such an approach is to balance model specificity with computational efficiency for use with optimization solvers.

MPCPy generates and solves dynamic optimization problems based on a user-provided Modelica model, objective, and constraint information in Python, and input time-series data (e.g., weather and electricity prices). The problems are solved using JModelica.org [72], which integrates CasADi [73] to compute function derivatives and IPOPT [74] as

Table 6
Configurations for the six different grid signals tested in this work using the same Model Predictive Control (MPC) framework.

DF tests	Objective	Grid signal	Constraints for power
Time-of-use tariff	Minimize cost of energy and demand	Energy and demand price	None
Real-time pricing	Minimize cost of energy	Dynamic energy price	None
Demand limiting	Minimize cost of energy and demand	Energy and demand price, power limit P_{lim}, start time t_s, final time t_f	$P \leq P_{lim}$ for $[t_s, t_f]$
Load shedding	Minimize cost of energy and demand	Energy and demand price, reference power P_{ref}, decreased power P_{dec}, start time t_s, final time t_f	$P \leq P_{ref} - P_{dec}$ for $[t_s, t_f]$
Load shifting	Minimize cost of energy and demand	Energy and demand price, reference power P_{ref}, decreased power P_{dec}, increased power P_{inc}, start increase time $t_{s,inc}$, final increase time $t_{f,inc}$, start decrease time $t_{s,dec}$, final decrease time $t_{f,dec}$	$P \geq P_{ref} + P_{inc}$ for $[t_{s,inc}, t_{f,inc}]$ and $P \leq P_{ref} - P_{dec}$ for $[t_{s,dec}, t_{f,dec}]$
Load tracking	Minimize error	Reference power P_{ref}, start time t_s, final time t_f	$\mid P - P_{ref} \mid < \epsilon$ for $[t_s, t_f]$

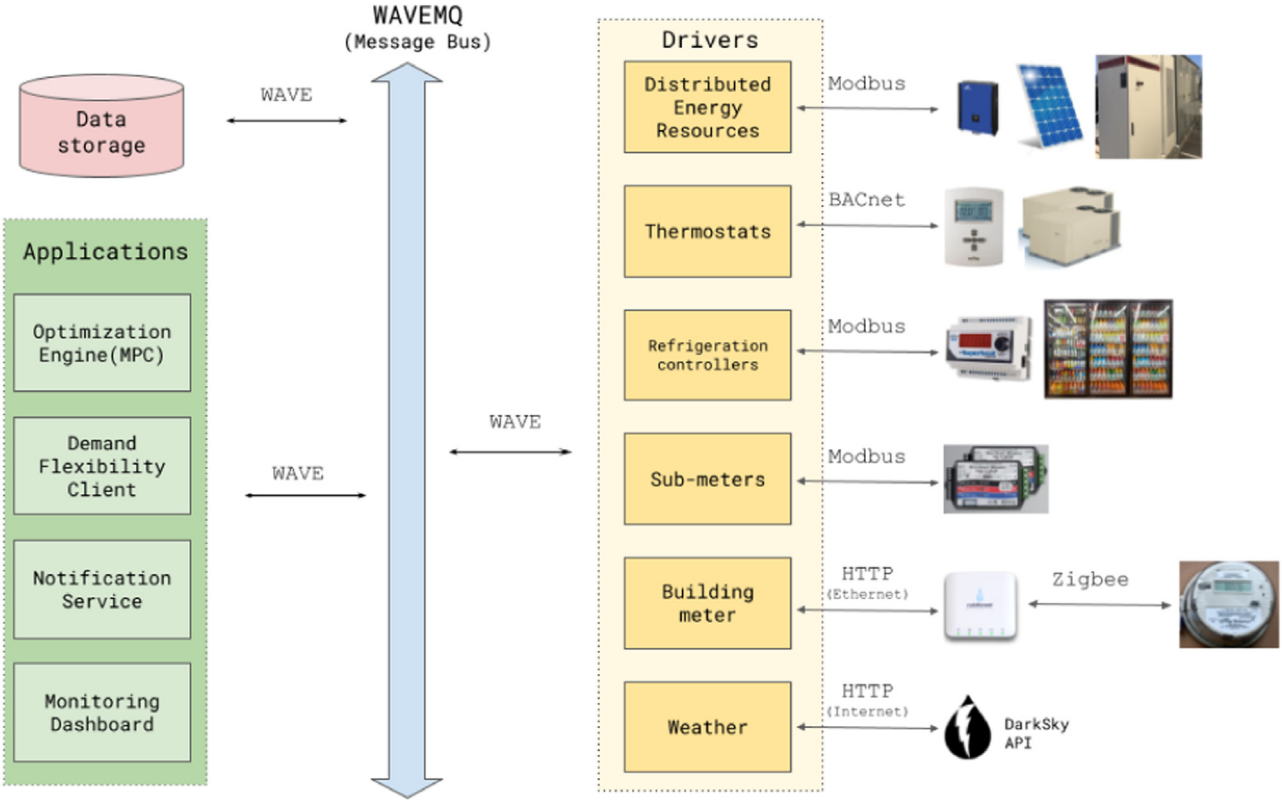


Fig. 7. The software architecture of the SolarPlus Optimizer (SPO) system: the applications and database exchange information with the drivers through WAVE and WAVEMQ; the drivers exchange information with the hardware and web service via their specific protocol.

the optimization algorithm. In MPCPy, the optimization problem is expressed in continuous time. It is passed into JModelica and discretized using the direct collocation method, which approximates the dynamic model variables using piecewise polynomials. This results in a large and sparse Nonlinear Program (NLP) solved by the algorithm IPOPT. IPOPT, short for interior point optimizer, is a state-of-the-art nonlinear optimization library, suitable to solve functions that are twice continuously differentiable. The first and the second derivatives are usually computed via algorithmic differentiation in modeling environments such as CasAdi, as is the case for JModelica. Once an NLP solution is converged, IPOPT postprocesses the solution to extract an approximate solution to the original dynamic optimization problem.

The software developed was hosted on a local computer, placed in the store. Data was also periodically pushed to the cloud for backup and analysis. Details about the remote communication and control, cloud data transfer, robustness to network outrages, and cybersecurity of the platform can be found in [13].

5. Field test results

5.1. Bill minimization

Based on the operational data, the regression model, and the evaluation method described in Section 3, we computed a monthly average electricity bill for the BAU and SPO modes for each season by scaling the electricity and demand charges to be representative of a real bill. Using this monthly average, we estimated the seasonal cost of both modes. Note that we excluded the operational data of event-based flexibility tests in the calculations. The left bar plots in Fig. 8 show both the energy and demand cost for the three seasons. We can see that SPO produced cost savings for all seasons compared to the BAU baseline.

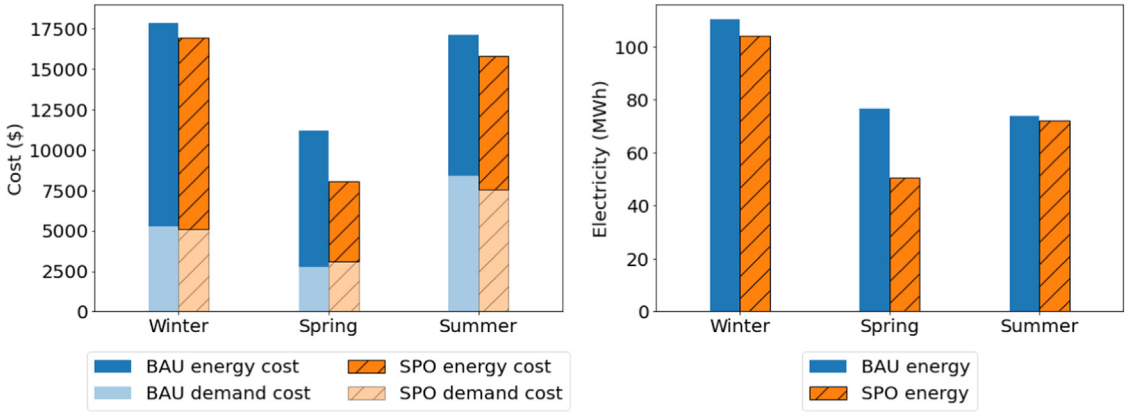


Fig. 8. Estimated seasonal utility bills (left) and energy use (right) in the two operation modes: Business as usual (BAU) and SolarPlus Optimizer (SPO). The annual electricity bill includes three seasons: winter, spring and summer.

Table 7
Average demand comparison of the two operation modes: Business As Usual (BAU) and SolarPlus Optimizer(SPO) across the whole year.

Season	Mode	Peak	Part Peak	Super Off/Off Peak
Winter	BAU (kW)	44	-	49
	SPO (kW)	23	-	48
Spring	BAU (kW)	40	-	41
	SPO (kW)	31	-	52
Summer	BAU (kW)	37	44	56
	SPO (kW)	25	48	55
Average	Demand reduction (%)	34	-9	-8

Overall, SPO saved 11.7% of the annual energy cost with an absolute savings of $5402/year.

Savings were mostly due to reductions in energy use (Fig. 8, right plots). Comparing the savings in the three seasons, we can see that SPO obtained higher absolute cost savings in spring compared to winter and summer. In spring, the SPO demand cost was slightly higher because SPO had a higher off-peak demand than BAU (52 vs. 41 kW), which can be observed in Table 7. SPO could however achieve a lower energy consumption and overall, a lower electricity bill. In winter and summer, SPO savings for energy and demand were both moderate. Due to a higher demand price in summer, we observe that the monthly demand cost weighed more than the energy cost for both operation modes, compared to the other two seasons. Table 7 summarizes the maximum demand for each season for the two operation modes. Note that the super-off-peak period has the same demand charge as the off-peak period in spring. We can see that SPO excelled at reducing peak demand, with an average reduction of 34% against BAU. On the other hand, SPO also resulted in 8%-9% higher demand during part-peak and off-peak periods.

Fig. 9 displays the total building demand under the two operation modes on a day of the three seasons. The shaded areas indicate either the peak, part-peak or super-off-peak period (see legends in each subplot), while the blank areas indicate the off-peak period. During the winter peak period, BAU tried to reduce the building demand but did not last for the whole six hours. On the other hand, SPO could maintain a lower demand consistently during the whole peak duration. This resulted in a lower demand cost in winter for SPO as discussed previously. In spring, the BAU operation mode was not responsive to the different rates of tariff during peak and super-off-peak periods, while SPO took advantage of the lower prices consistently as in the other two seasons. This explains the energy savings for SPO in spring against the BAU mode.

Among all the controlled energy systems, the battery influences the overall building power profile the most due to its large flexibility potential compared to the other systems (see Table 2). Fig. 10 shows the daily battery power profiles of the SPO and BAU modes averaged over the summer season. We can see that both modes were able to adjust charging and discharging power in response to the different tariff rates. SPO however shows a more consistent behavior, especially during the peak period. As a result, SPO could reduce the overall demand costs as well as energy costs.

5.2. Flexibility events

5.2.1. Real-time pricing and demand limiting

Table 8 summarizes the test results of the five types of DF events. The RTP tests show 3% of energy cost savings of SPO against the reference scenario, which does not respond to RTP. The savings were positive, but not significant. The tested RTP signal [13] is an hourly-averaged curve with smooth changes from 0.08 to 0.18 $/kWh. For about 63% of the time, the prices fall in the middle range between 0.10 to 0.15 $/kWh. The high price (higher than 0.15 $/kWh) lasts only 3 hours and the low price (lower than 0.10 $/kWh) lasts 6 hours. During that middle price range, it was not advantageous for MPC to react to the continuous changes to minimize cost. The MPC strategy thus used less energy when the energy price was high and more energy when the price was low. Compared with the reference scenario, MPC increased the energy cost during the low price and decreased the cost during the high price. Overall, MPC increased energy use but still reduced the energy cost. It was expected that MPC would present more cost savings for buildings in response to RTP with a longer duration of high price, or with a larger discrepancy between the low and high prices.

In the demand limiting test, we set the building net load to be lower than 26 kW during the event, as described in Section 3.3. The test result shows 100% of satisfaction with the requirement. For simplicity, the results of these two tests are not detailed in this paper, as similar results have been presented in our previous paper [13].

5.2.2. Load shedding

Fig. 11 shows the results of the load shedding test. The red shaded area shows the shedding period from 15:00 to 18:00, which falls on the second half of the peak period from 12:00 to 18:00. The light red areas from 9:00 to 12:00 and from 18:00 to 21:00 are part-peak periods. We can see that SPO reduces the overall demand during the event and during the pre-event peak period (Fig. 11, first panel). SPO achieves these results by discharging the battery and, at the same time, relaxing the RTUs setpoints. The battery starts discharging at the start of the peak period (Fig. 11, second panel). Halfway through the load shedding event, however, the battery reaches the lower limit of SOC, a constraint set for this test as described in Section 3.3, and it starts idling until the end of the event. At the same time, SPO begins to increase the room temperature setpoints during the part-peak period and further increases up to the specified maximum during the load shedding event (shown by

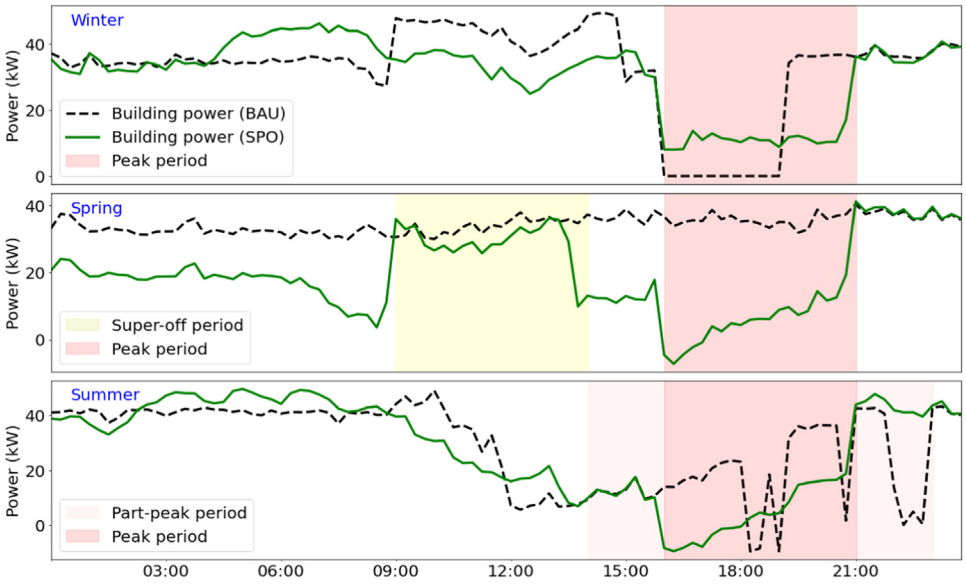


Fig. 9. Net building demand on a typical day of the three seasons in the two operation modes: Business as usual (BAU) and SolarPlus Optimizer (SPO).

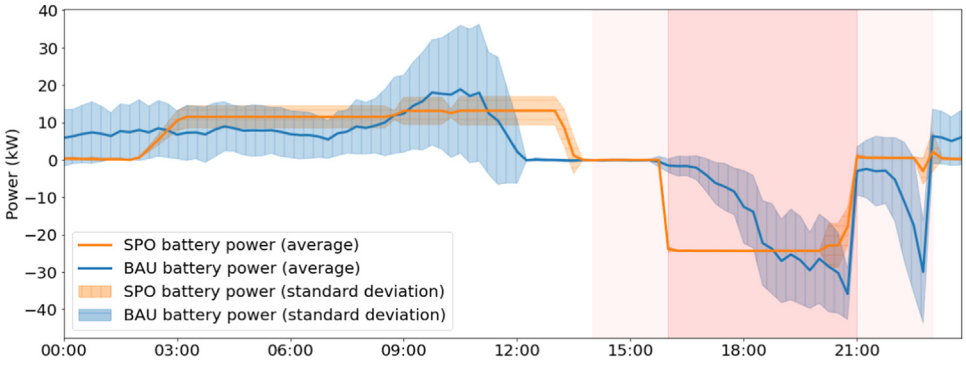


Fig. 10. Battery charge and discharge power comparison between the two operation modes: SolarPlus Optimizer (SPO) and Business as usual (BAU). The latter used the proprietary algorithm over the summer season.

Table 8
Test results of the five types of demand flexibility events using the SolarPlus Optimizer (SPO) operation mode.

Flexibility event	Primary objective	Key performance indicator (KPI)	Result
Real-time pricing	Minimize energy cost	Cost savings (%)	3%
Demand limiting	Keep load below 26 kW during 15:00–18:00	Time below demand limit (%)	100%
Load shedding	Reduce load by 3 kW during 15:00–18:00	Time of meeting requirement (%)	83%
Load	Increase load by 10 kW during 12:00–15:00	Time of meeting requirement (%)	100%
shifting	Decrease load by 10 kW during 15:00–18:00	Time of meeting requirement (%)	100%
Load tracking	Follow reference power signal	Time of meeting requirement (%)	56%

the red curve in 11, third panel). As a result, the RTU demand decreases, compared to the BAU strategy (shown by the orange dashed curve in the second panel of Fig. 11). After the event, the battery starts charging again and the setpoints go back to the baseline values. This test result shows that SPO can coordinate the controlled systems to achieve the test goal, by reducing HVAC power consumption and trading off comfort for cost savings, when the battery is constrained. Overall, by combining the HVAC and battery control, SPO was able to reduce the load by the specified amount 83% of the time.

5.2.3. Load shifting

Fig. 12 shows the test result for the load shifting event. SPO successfully increased the building demand during the whole load-increase period and reduced the building demand during the whole load-decrease period, compared to the BAU baseline (top panel). The lower plot of Fig. 12 shows the power difference between the SPO and BAU modes. We can see that the shifted amount was on average 20 kW. In this test, SPO achieved the goal by primarily controlling the battery. It discharged the battery just before the load increase period (Fig. 12, first panel in blue), to increase the amount of energy that could be stored during the load-increase period. During the load-decrease period, it kept discharging until the end of the test.

5.2.4. Load tracking

The blue curve in the upper plot of Fig. 13 shows the reference power profile (in blue) and the allowable ±3 kW error band around it (shaded area). The load tracking event lasts 24 h. The green curve in the upper panel of Fig. 13 shows the building power using the SPO strategy, while the dotted black line represents the BAU strategy. SPO was able to follow the profile more closely than BAU, although it was only able to track the signal within the error band for 56% of the time. In comparison to BAU, SPO reduced the signal violations by 35%. SPO achieved this result primarily by charging and discharging the battery, as displayed by the blue area in the lower panel of Fig. 13.

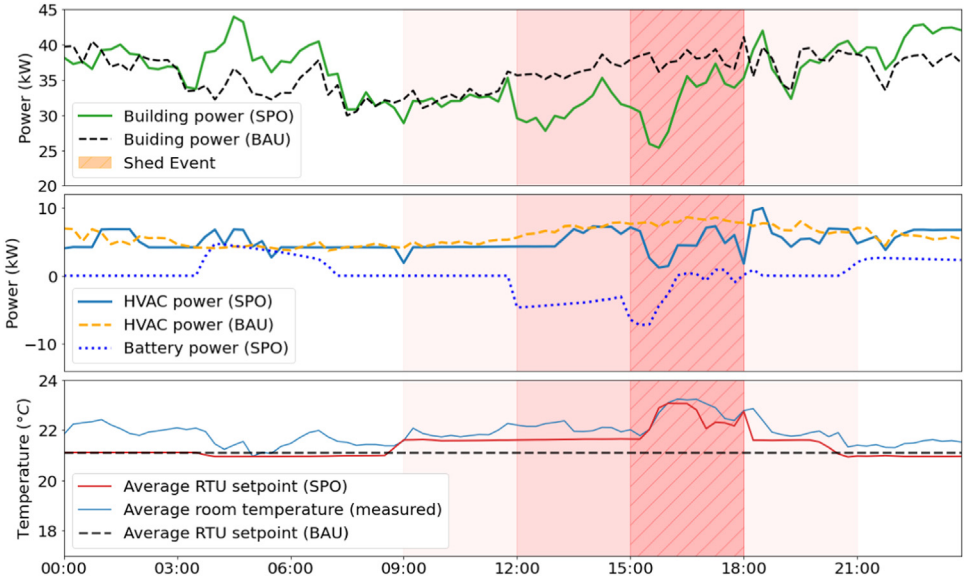


Fig. 11. Net building demand profile during a load shedding event in the SolarPlus Optimizer (SPO) operation mode on September 3, 2020, compared with the Business-As-Usual (BAU) mode.

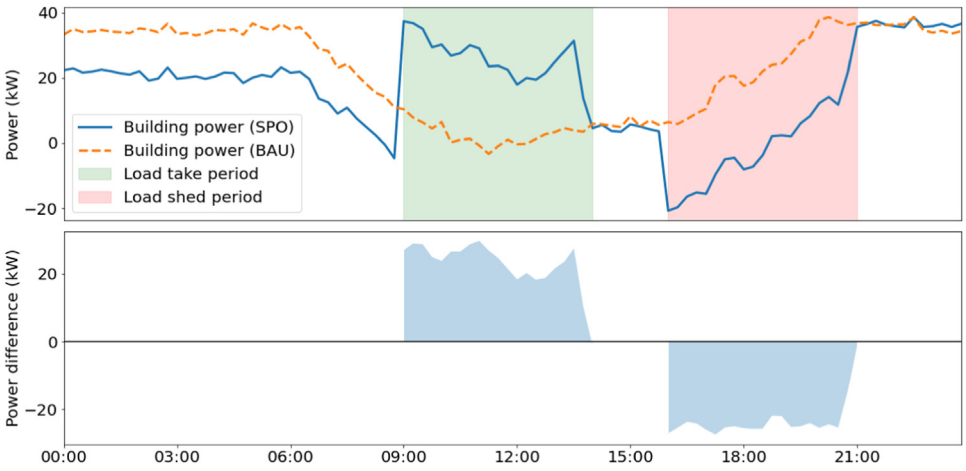


Fig. 12. Net building demand profile during a load shifting event in the SolarPlus Optimizer (SPO) operation mode on May 4, 2021, compared with the Business-As-Usual (BAU) mode.

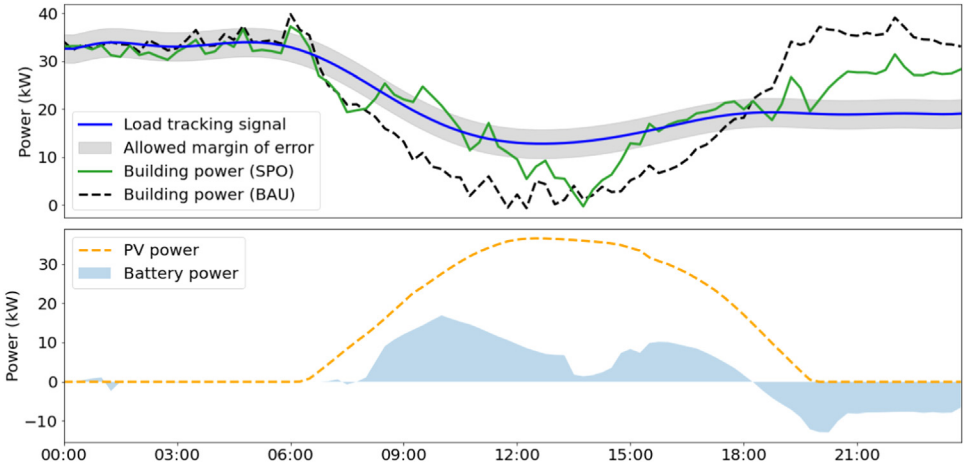


Fig. 13. Net building demand profile during a load tracking event in the SolarPlus Optimizer (SPO) operation mode on April 28, 2021, compared with the Business As Usual (BAU) mode.

6. Discussion

While the real-world operation using the SPO platform were overall positive and encouraging, they brought to light several important considerations.

6.1. Baseline and modeling uncertainties

In general, SPO showed a consistent behavior in response to the different tariffs across the three seasons, as presented in Section 5. SPO used less energy during peak and part peak periods and more during off-peak periods than BAU. SPO also showed lower demand in all peak periods and slightly increased the demand in part-peak or off-peak periods in comparison to BAU. The BAU power profile, determined largely by the battery controller, was unpredictable at times. For instance, we noticed its unresponsiveness to the spring tariff and sudden decreases and increases of demand during the summer peak and part-peak periods during certain evenings (see Fig. 9). We did not receive any explanation from the battery company that also provides the software service, but we guess that during the spring period, the controller was still "learning" how to optimize the system. These variations add uncertainty to the overall performance evaluation presented here. Future work should aim at reducing such uncertainty with additional testing under more controlled baseline behavior.

Among the five flexibility event tests, the RTP test is similar to the bill minimization using the TOU tariff, because both operations involve only cost minimization and no constraints on the demand. The rest four flexibility event tests, on the other hand, include an extra constraint on the building net demand. To better perform in these tests, a more precise forecast of the building demand and PV generation is required. With better model accuracy, it is expected that the RTP test would result in higher cost savings. The building would also follow the reference load better for the load tracking event, which has a more stringent requirement (with an error margin of ±3 kW or 7% of the peak demand). The latter four events also require more accurate coordination and sequencing of the controlled energy systems, as the on/off state of these systems will determine directly whether the total demand falls within the constraints or not.

The building thermal model in the SPO supervisory controller was used to predict future temperatures of the thermal zones subject to controlled and uncontrolled inputs. The model was therefore calibrated for temperatures only, as shown in Fig. 4. To achieve better prediction of the building power demand and better DF performance, a calibration of the power demand of the building thermal model could be helpful.

6.2. Selection of controlled systems

A large part of the SPO cost savings and DF were achieved by operating the battery. This was due to the flexibility potential of the battery, compared to the other systems. At the test site, the HVAC demand was only 15–25% of the total building load, due to the large share of the load caused by the uncontrollable gaming machines (see Section 3.1). It is expected that a commercial building with a larger share of HVAC loads will have a higher potential for energy savings and flexibility.

Further, the refrigerator and freezer hardly contributed to the cost savings and DF of the system. The principal reason is that these two systems were already operating to their physical limits (i.e., they are undersized) in the BAU mode. Setting the refrigerator and freezer setpoints to their maximum allowed bounds (0.5 °C increase for the refrigerator and 3 °C increase for the freezer) in the SPO mode did not produce any noticeable energy reduction. The facility operator did not recommend increasing their setpoints to higher limits due to food safety concerns. In future work, we suggest conducting a qualitative evaluation of the load flexibility potential of the investigated energy systems before including them in the control design plan.

6.3. Cost vs. more granular sensing

We designed the SPO system to be scalable and low-cost. For this reason, we installed only two thermostats, two refrigeration controllers, and five sub-meters, which measure temperatures and powers for the two RTUs, the refrigerator, the freezer, and the battery system. We did not install any additional sensors to measure the temperature distribution in each zone. However, it is likely that temperature sensors onboard the thermostats, located as in Fig. 1, did not capture gradients in the spaces, especially at the front desk. The staff often required temperature adjustments in that zone, and it was often impossible to aggressively lower the cooling setpoints given the lack of accurate temperature measurement to estimate thermal comfort. This represented an additional limitation to the HVAC energy savings and flexibility.

Furthermore, the lack of measurements of the supply airflow and temperature limited the estimation of heat extracted or added to the space by the RTUs. Internal heat gains from the lighting and plug loads were also unmeasured, making it difficult to accurately model the dynamics of the thermal zones. Future work should consider the trade-off between cost and granularity of sensing, especially with the increased availability of inexpensive sensors. In general, we always recommend adding necessary temperature sensors to better assess the thermal comfort of occupants. Additional sensing may also provide more reliable demand flexibility services to the grid.

For the freezer, the defrost cycle used a heating element with a non-negligible nominal power of the same magnitude as its cooling power, unlike the refrigerator that used air defrost. The heating element was wired separately in the system, and therefore our installed power meter did not measure the heating power during the defrost cycle. This made controlling the defrost cycle difficult. In future work, the power of the defrost cycle for freezer applications should be more carefully considered.

6.4. Modeling choices

The two RTU systems are controlled using room temperature setpoints in our SPO implementation. The operation results showed that this control strategy could not precisely control the on/off cycles of the RTUs and therefore could not avoid the stacked peak demand associated with simultaneous equipment operations. For better equipment scheduling, mixed-integer linear programming should be used to solve this optimization problem, or an additional coordination layer on top of the current SPO structure should be used to mitigate this control problem.

This work adopted a simple RC network to model the building thermal behavior. A systematic approach could be used to identify the model structure with the optimal number of resistances and capacitances for the studied building [75]. To improve the model accuracy for prediction and forecasting, operational data could be used to train, validate and test the identified model in various scenarios. Experiments could also be conducted to obtain larger temperature oscillations in the thermal zones to achieve sufficient system excitation for better system identification results. Better forecasting models for internal heat gains could also be used, for instance, using black-box modeling approaches [41] or an improved disturbance model [33]. We adopted a deterministic MPC structure in this work. To systematically accommodate uncertainties associated with model prediction, forecasting and measurement errors, a robust or stochastic MPC structure [76] will be considered in future work.

6.5. Large-scale implementation

The SPO platform was designed to be scalable and adaptable to new buildings. It is vendor- and protocol-agnostic and particularly suitable for SMBs without a BAS installed. It applies to any such SMBs around the world, even though it was only deployed and tested in a building located in the U.S. The MPC formulation is also flexible to handle various

energy cost minimization and DF scenarios. However, to scale up the implementation, we need to overcome several obstacles: (1) a process to identify the minimum sensing and measuring devices for a variety of systems and building types; (2) a method to rapidly obtain a simple yet accurate controller model, (3) a robust MPC framework which is able to adapt to various uncertainties and operational changes; (4) an integrated library of optimization solvers to handle a range of optimization problems.

7. Conclusion

In this paper, we presented a software platform called "SolarPlus Optimizer (SPO)". The platform primarily includes two modules: a middleware that acts as a building automation system and an optimization engine using Model Predictive Control (MPC). We built the software exclusively on open-source libraries and designed it to be scalable and extensible. We implemented and deployed the platform on a convenience store and gas station in Northern California. We also described the flexible optimization framework, which was able to handle different operating modes with the same formulation. The controlled energy systems in this small commercial building included space conditioning, a refrigerator, a freezer, and an electrochemical battery. We tested two types of demand flexibility applications: electricity bill minimization using the time-of-use tariff and responses to flexibility events. We operated the SPO and the baseline (BAU) modes for controlling those energy systems for a year. The long-term operation results showed that the optimization engine saved 12% of the annual energy cost against the BAU operation. The average peak reduction for MPC is 34%. We also conducted tests on the building for demand flexibility using five different types of events: real-time pricing, demand limiting, load shedding, load shifting, and load tracking. The test results showed that MPC could well adjust the building demand to achieve the specific goals in those tests.

8. Software availability

The source code for the SPO platform as well as the models used in this work are available at https://github.com/LBNL-ETA/SolarPlus-Optimizer. Instructions on how to install and use the software have been detailed on the website.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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